

Capture the Flag 2: Widgets

Goal: Turn the single-card application into a simple multi-section interface.

Note from the instructor before you start:

These activities involve using your own problem solving and researching skills. We are strictly to implement NOT TO USE AI to generate the code for you. You may only use AI for RESEARCHING the syntax or usage, but NEVER THE CODE ITSELF.

Consequence: Immediate 0 score in your grade for this activity.

(I know what AI-generated code looks like, don't even think about it :>)

🚩 Flag 0 – Create Your Feature Branch

Objective: Create a feature branch before modifying your application. Name the branch feature/widgets

Screenshot:

- Terminal showing `feature/widgets`

```
C:\Windows\System32\cmd.e x + v
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\johnf\Downloads\ggs>git branch
* main

C:\Users\johnf\Downloads\ggs>git checkout -b feature/widgets
Switched to a new branch 'feature/widgets'

C:\Users\johnf\Downloads\ggs>git branch
* feature/widgets
  main

C:\Users\johnf\Downloads\ggs>|
```

🚩 Flag 1 – Break It Apart

Objective: Separate your profile information into individual `Text` widgets.

If your card currently looks like this:

```
Text(
  'Chiara Marie C. Canque\n'
  'BSIT - 3\n'
  'My First Flutter Application\n'
  'August 11, 2026',
)
```

Your task: Separate each piece of information into its own `Text` widget.

You should have something conceptually like:

Text → Name

Text → Course & Section

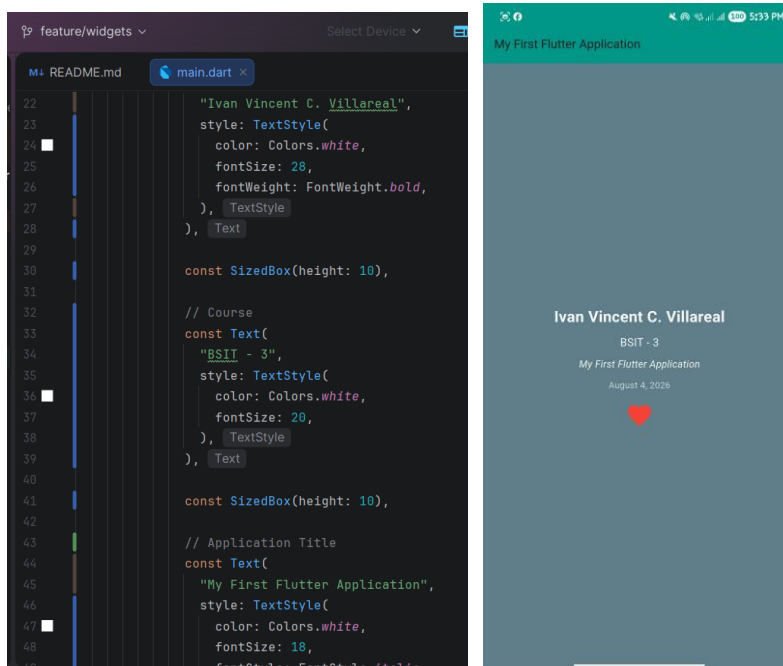
Text → Application Title

Text → Date

Do not worry about making it pretty yet.

Screenshot:

- Your code showing the separate `Text` widgets
- Your application



🚩 Flag 2 – Stack the Information

Objective: Arrange your individual `Text` widgets vertically.

Research how to use:

`Column()`

Place your information inside the `Column`.

Your card should now have:

Name

Course & Section

Application Title

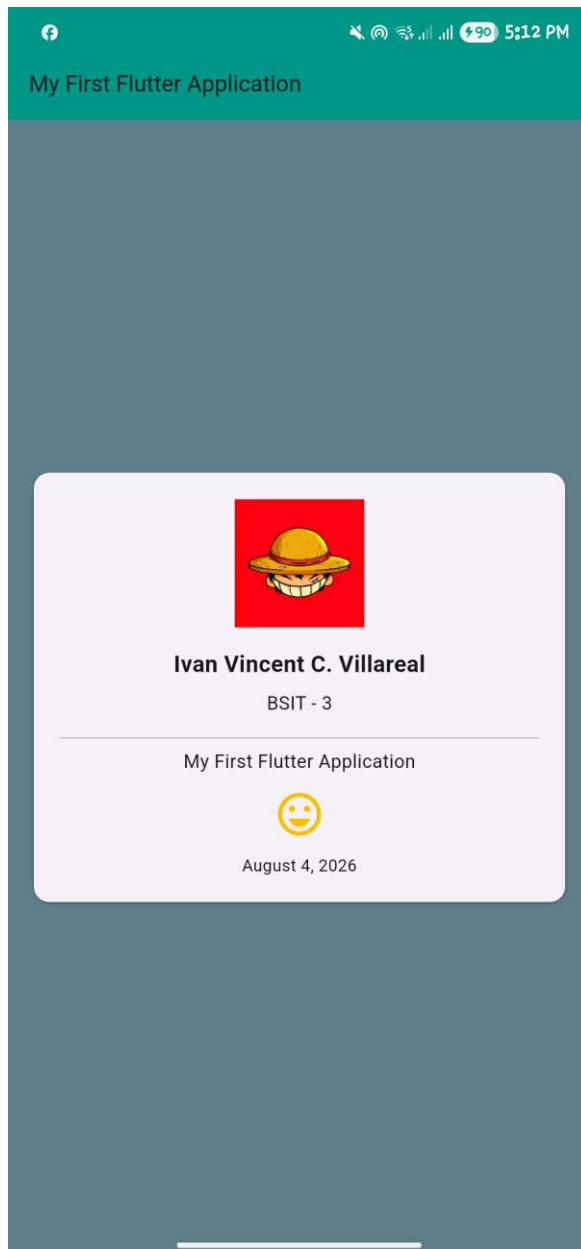
Date

Screenshot:

- The `Column` in your code
- The four separate pieces of information displayed vertically

```
main.dart
11
12 margin: const EdgeInsets.all(20),
13 child: Padding(
14   padding: const EdgeInsets.all(20),
15   child: Column(
16     mainAxisAlignment: MainAxisAlignment.min,
17     children: [
18       Image.asset(
19         "assets/download.jpg", // Change this if your filename is different
20         width: 100,
21         height: 100,
22       ), Image.asset
23
24       const SizedBox(height: 15),
25
26       const Text(
27         "Ivan Vincent C. Villareal",
28         style: TextStyle(fontSize: 22, fontWeight: FontWeight.bold),
29       ), Text
30
31       const SizedBox(height: 8),
32
33       const Text("BSIT - 3", style: TextStyle(fontSize: 18)),
34
35       const SizedBox(height: 8),
36
37       const Divider(),
38
```

```
main.dart
49
50 const Text(
51   "My First Flutter Application",
52   style: TextStyle(fontSize: 18),
53 ), Text
54
55 const SizedBox(height: 10),
56
57 const Icon(
58   Icons.sentiment_very_satisfied,
59   color: Colors.amber,
60   size: 40,
61 ), Icon
62
63 const SizedBox(height: 10),
64
65 const Text("August 4, 2026", style: TextStyle(fontSize: 16)),
66 ],
67 ), Column
68 ), Padding
69 ), Card
70 ), Center
71 ), Scaffold
72 ); MaterialApp
73 }
74
```



🚩 Flag 3 – Add Another Element

Objective: Add another individual widget to your `Column`.

Add something that belongs to your profile.

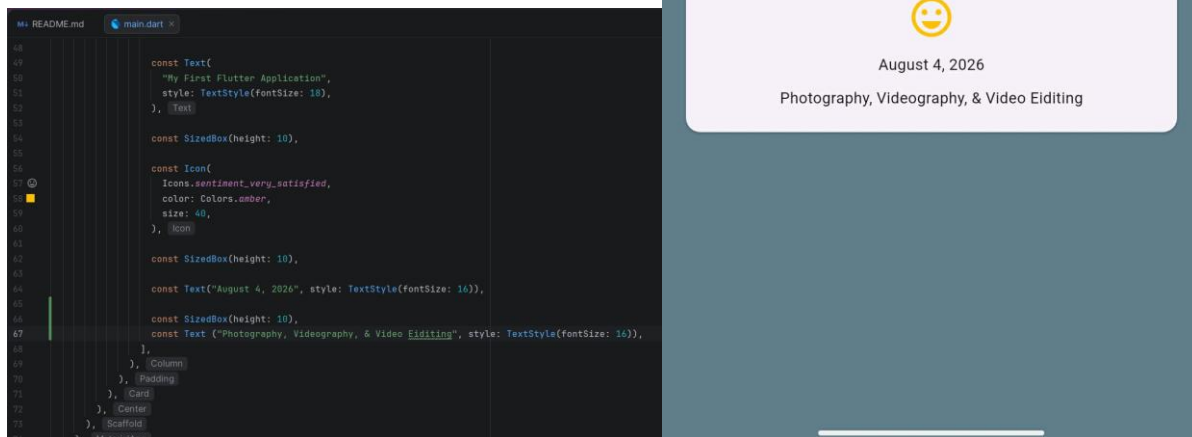
Examples:

Hobby

It must be its **own widget**, not additional text inside another `Text`.

Screenshot:

- New widget in your `Column`
- Updated application



🚩 Flag 4 – Put Two Things Side-by-Side

Objective: Discover the `Row` widget.

Create a new section containing **two pieces of information beside each other**.

For example:

Age: 21 Birthdate: August 11, 2003

Research:

`Row()`

Place the `Row` inside your existing `Column`.

Your structure should now resemble:

Column

├─ Text

├─ Text

├─ Text

├─ Text

├─ Text

└─ Row

 ├─ Text

 └─ Text

Screenshot:

- Your widget structure/code
- The two items displayed side-by-side

```
M+ README.md main.dart x
25
26 child: Padding(
27   padding: const EdgeInsets.all(20),
28
29   child: Column(
30     mainAxisAlignment: MainAxisAlignment.min,
31
32     children: [
33       Image.asset(
34         "assets/download.jpg",
35         width: 100,
36         height: 100,
37       ), Imageasset
38
39       const SizedBox(height: 15),
40
41       const Text(
42         "Ivan Vincent C. Villareal",
43         style: TextStyle(
44           fontSize: 22,
45           fontWeight: FontWeight.bold,
46         ), TextStyle
47     ), Text
48
49     const SizedBox(height: 8),
50
51     const Text(
```

```
M+ README.md main.dart x
49 const SizedBox(height: 8),
50
51 const Text(
52   "BSIT - 3",
53   style: TextStyle(fontSize: 18),
54 ), Text
55
56 const SizedBox(height: 8),
57
58 const Divider(),
59
60 const Text(
61   "My First Flutter Application",
62   style: TextStyle(fontSize: 18),
63 ), Text
64
65 const SizedBox(height: 10),
66
67 const Icon(
68   Icons.sentiment_very_satisfied,
69   color: Colors.amber,
70   size: 40,
71 ), Icon
72
73 const SizedBox(height: 10),
74
75 const Text(
```

```
104
105
106 // Hobby - Flag 3
107 const Text(
108   "Photography, Videography, & Video Editing",
109   style: TextStyle(fontSize: 16),
110 ), Text
111
112 const SizedBox(height: 15),
113
114 // Row - Flag 4
115 const Row(
116   mainAxisAlignment: MainAxisAlignment.center,
117   children: [
118     Text(
119       "Age: 24",
120       style: TextStyle(fontSize: 16),
121     ), Text
122
123     SizedBox(width: 30),
124
125     Text(
126       "Birthdate: January 2002",
127       style: TextStyle(fontSize: 16),
128     ), Text
129   ], Row
130 ),
131 ],
```



🚩 Flag 5 – Give It Some Space

Objective: Improve the spacing between your widgets.

Research:

`SizeBox()`

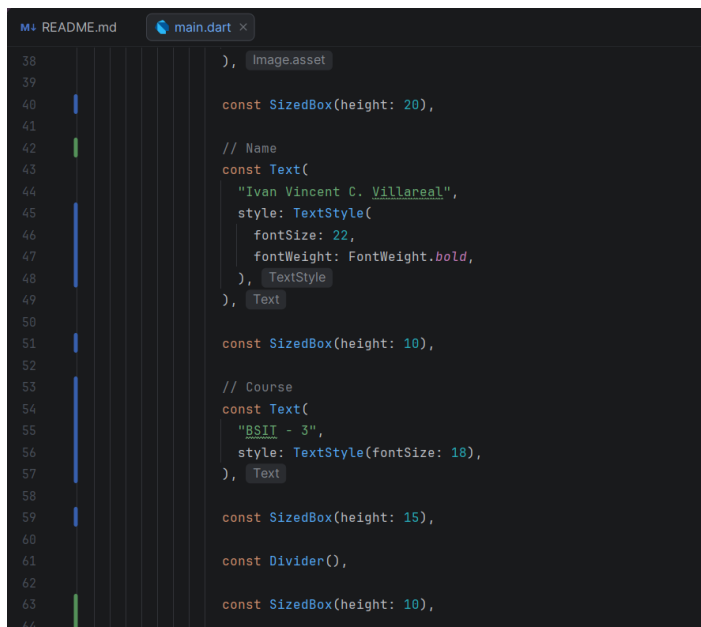
Add appropriate spacing between your:

- Image
- Name
- Course
- Other information
- Row

Challenge: Make the card easier to read **without changing its colors or fonts yet.**

Screenshot:

- `SizeBox` usage
- Updated application



```
38      ), image.asset
39
40      const SizedBox(height: 20),
41
42      // Name
43      const Text(
44        "Ivan Vincent C. Villareal",
45        style: TextStyle(
46          fontSize: 22,
47          fontWeight: FontWeight.bold,
48        ),
49      ),
50
51      const SizedBox(height: 10),
52
53      // Course
54      const Text(
55        "BSIT - 3",
56        style: TextStyle(fontSize: 18),
57      ),
58
59      const SizedBox(height: 15),
60
61      const Divider(),
62
63      const SizedBox(height: 10),
64
```

```

M+ README.md  main.dart x
65 // Application Title
66 const Text(
67   "My First Flutter Application",
68   style: TextStyle(fontSize: 18),
69 ), Text
70
71 const SizedBox(height: 15),
72
73 // Icon
74 const Icon(
75   Icons.sentiment_very_satisfied,
76   color: Colors.amber,
77   size: 40,
78 ), Icon
79
80 const SizedBox(height: 15),
81
82 // Date
83 const Text(
84   "August 4, 2026",
85   style: TextStyle(fontSize: 16),
86 ), Text
87
88 const SizedBox(height: 10),

```

```

M+ README.md  main.dart x
89
90 // Hobby
91 const Text(
92   "Photography, Videography, & Video Editing",
93   style: TextStyle(fontSize: 16),
94 ), Text
95
96 const SizedBox(height: 15),
97
98 // Flag 4 - Row
99 const Row(
100   mainAxisAlignment: MainAxisAlignment.center,
101   children: [
102     Text(
103       "Age: 24",
104       style: TextStyle(fontSize: 16),
105     ), Text
106
107     SizedBox(width: 30),
108
109     Text(
110       "Birthdate: June 2002",
111       style: TextStyle(fontSize: 16),
112     ), Text
113   ],
114 ), Row
115
1

```



🚩 Flag 6 – Make It Look Better

Objective: Apply styling to your individual widgets.

Now that your information is properly separated and positioned, improve the visual hierarchy.

Research how to modify:

`TextStyle()`

Apply different styling to at least **three** elements.

For example:

- Name → larger and bold
- Course → medium size
- Date → smaller
- Application title → italic

Screenshot:

- Your `TextStyle` changes
- Updated application

```

41
42 // Name
43 const Text(
44   "Ivan Vincent C. Villareal",
45   style: TextStyle(
46     fontSize: 26,
47     fontWeight: FontWeight.bold,
48   ),
49 ),
50
51 const SizedBox(height: 10),
52
53 // Course
54 const Text(
55   "BSIT - 3",
56   style: TextStyle(
57     fontSize: 20,
58   ),
59 ),
60
61 const SizedBox(height: 15),
62
63 const Divider(),
64
65 const SizedBox(height: 10),
66
67 // Application Title
68 const Text(
69   "My First Flutter Application",

```



🚩 Flag 7 – Improve the Layout

Objective: Control the alignment of your widgets.

Research the properties of `Column` and `Row`.

Experiment with:

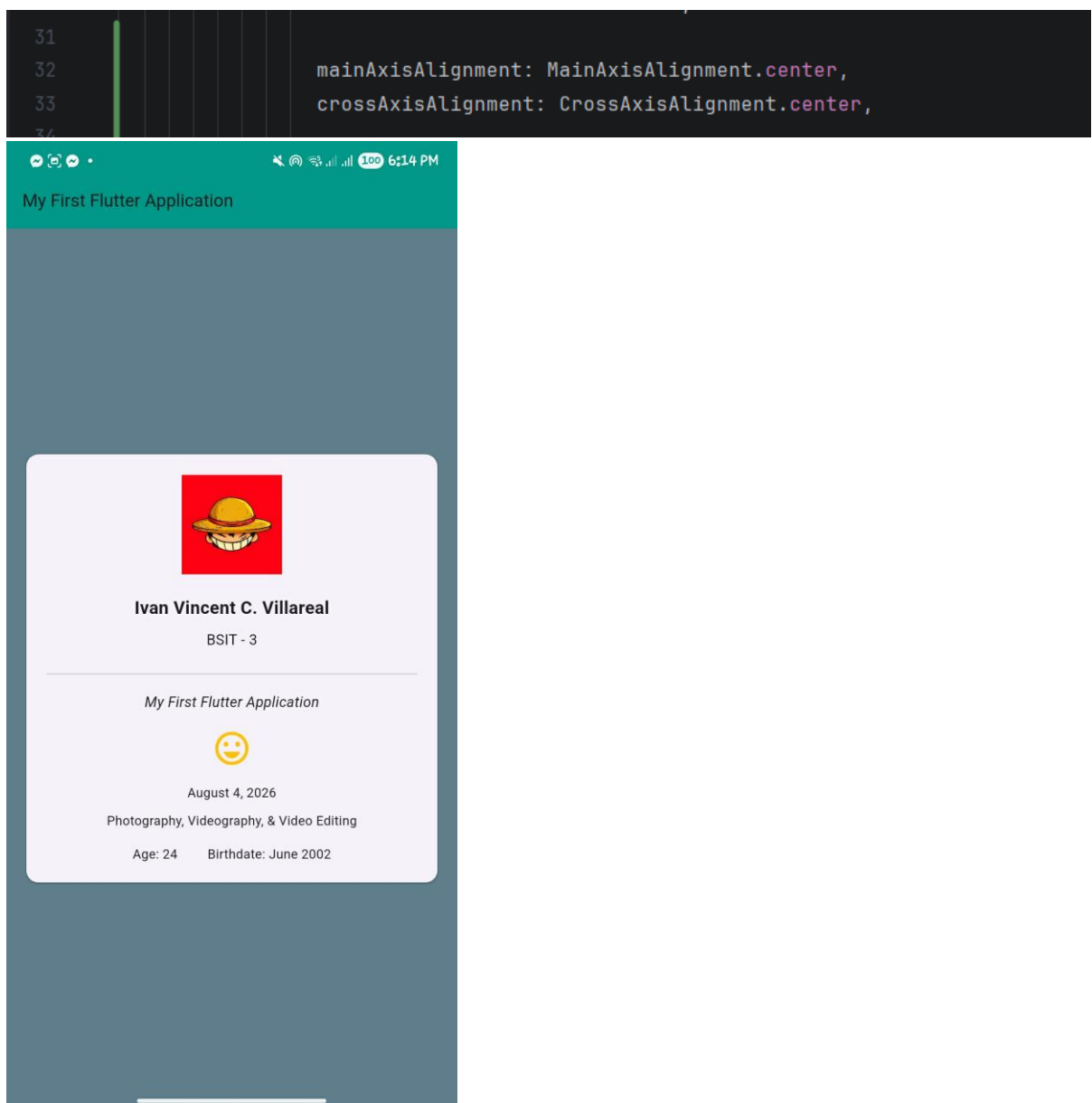
`mainAxisAlignment`

`crossAxisAlignment`

Change the alignment of your content so the card looks intentional rather than simply stacked.

Screenshot:

- The properties you changed
- Updated application



🚩 Flag 8 – Create a Second Card

Objective: Apply what you learned to create another section.

Create a second **Card** containing something different from your profile.

For example:

My Favorites

Favorite Game

Favorite Food

Favorite Movie

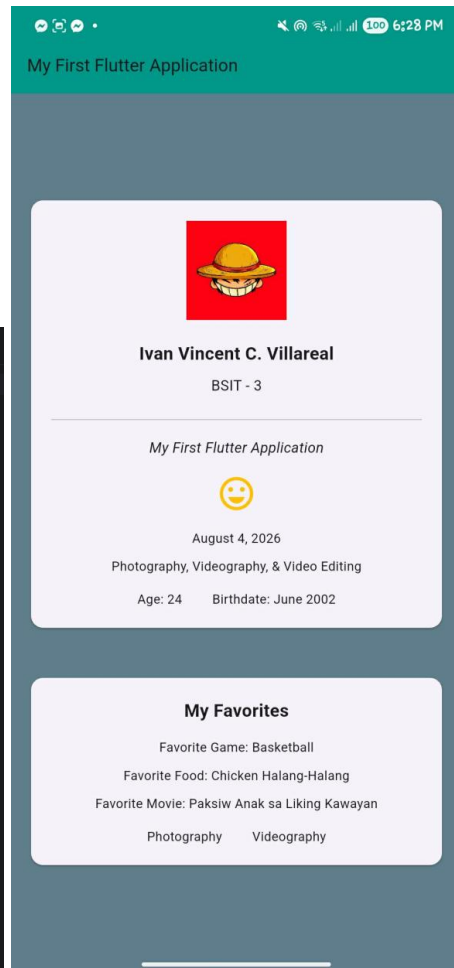
The second card should also use:

- Individual **Text** widgets
- A **Column**
- At least one **Row**
- Spacing

Screenshot:

- Code for the second card
- Both cards displayed

```
M+ README.md main.dart x
141 //FLAG 8
142 Card(
143   margin: const EdgeInsets.all(20),
144
145   child: Padding(
146     padding: const EdgeInsets.all(20),
147
148     child: Column(
149       mainAxisAlignment: MainAxisAlignment.min,
150       mainAxisAlignment: MainAxisAlignment.center,
151       crossAxisAlignment: CrossAxisAlignment.center,
152
153       children: [
154
155         const Text(
156           "My Favorites",
157           style: TextStyle(
158             fontSize: 22,
159             fontWeight: FontWeight.bold,
160           ), Text
161         ), Text
162
163         const SizedBox(height: 15),
164
165         const Text(
166           "Favorite Game: Basketball",
167           style: TextStyle(fontSize: 16),
168         ), Text
```



🚩 Flag 9 – Stack the Cards

Objective: Arrange your two cards vertically.

You now have:

Profile Card



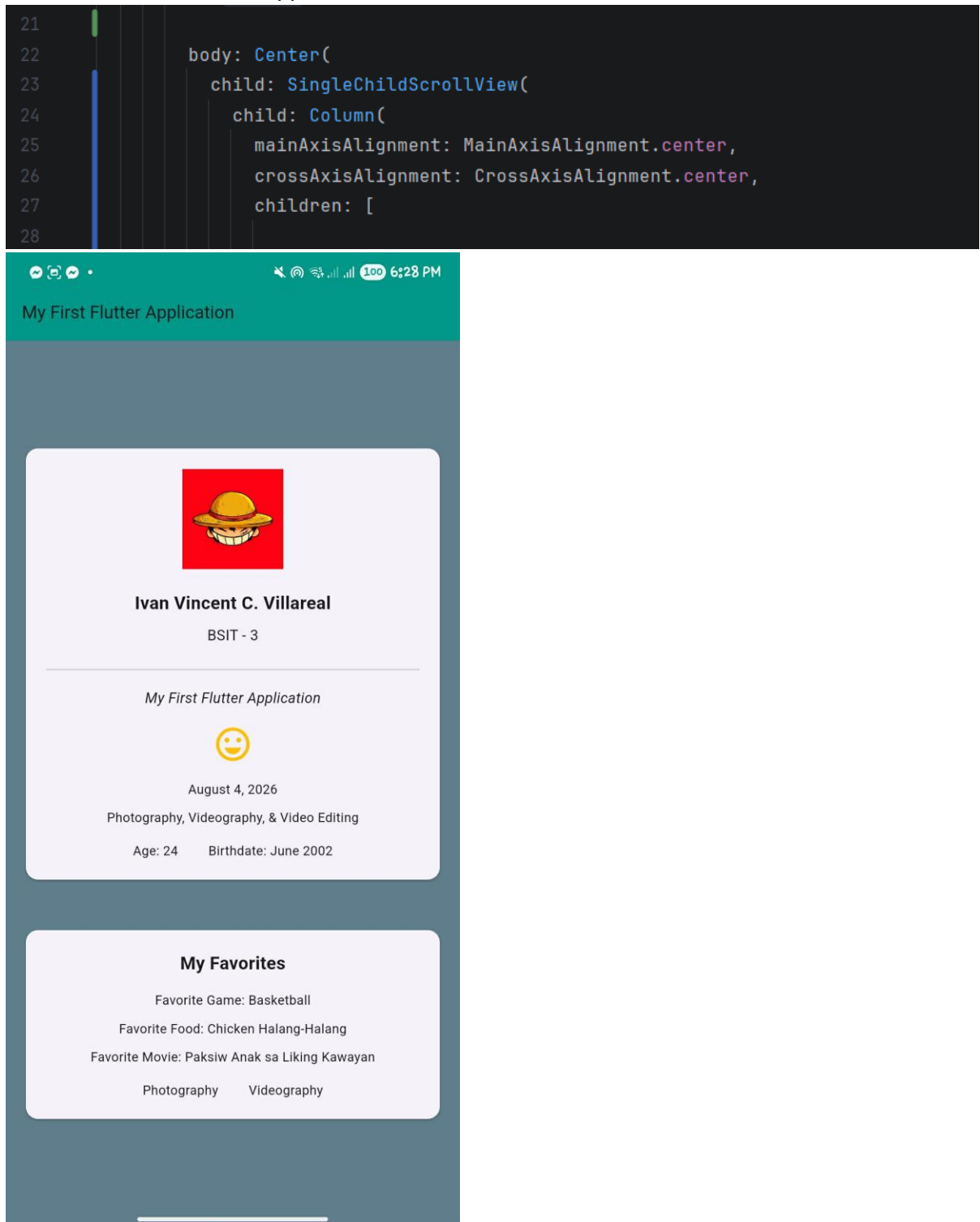
Favorites Card

Research how to place **multiple widgets inside the Flutter body**.

Use a layout widget to arrange your two cards vertically.

Screenshot:

- The widget containing both Cards
- Both cards visible in the application



🚩 Flag 10 – Widget Discovery

Objective: Discover a Flutter widget independently.

Research the Flutter widget catalog and find **one widget you haven't used yet**.

Examples:

- `Icon`
- `Divider`
- `CircleAvatar`
- `Chip`
- `ElevatedButton`

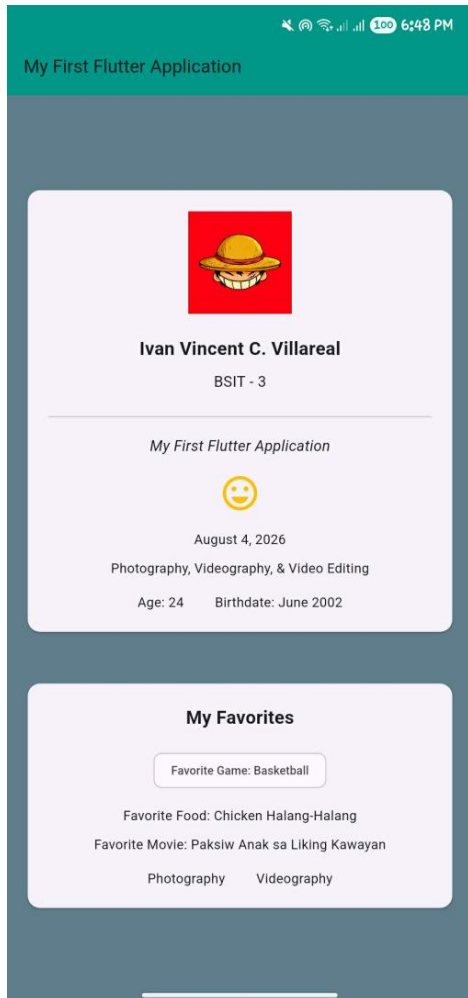
Add it somewhere appropriate in your application.

Screenshot:

- Your newly discovered widget in the code

- The widget visible in the application

```
162      const Chip(  
163        label: Text("Favorite Game: Basketball"),  
164      ), Chip  
165  
166      const SizedBox(height: 10),  
167  
168      const Text(  
169        "Favorite Food: Chicken Halang-Halang",  
170        style: TextStyle(fontSize: 16),  
171      ), Text  
172  
173      const SizedBox(height: 10),  
174  
175      const Text(  
176        "Favorite Movie: Paksiw Anak sa Liking Kawayan",  
177        style: TextStyle(fontSize: 16),  
178      ), Text  
179  
180      const SizedBox(height: 15),  
181
```



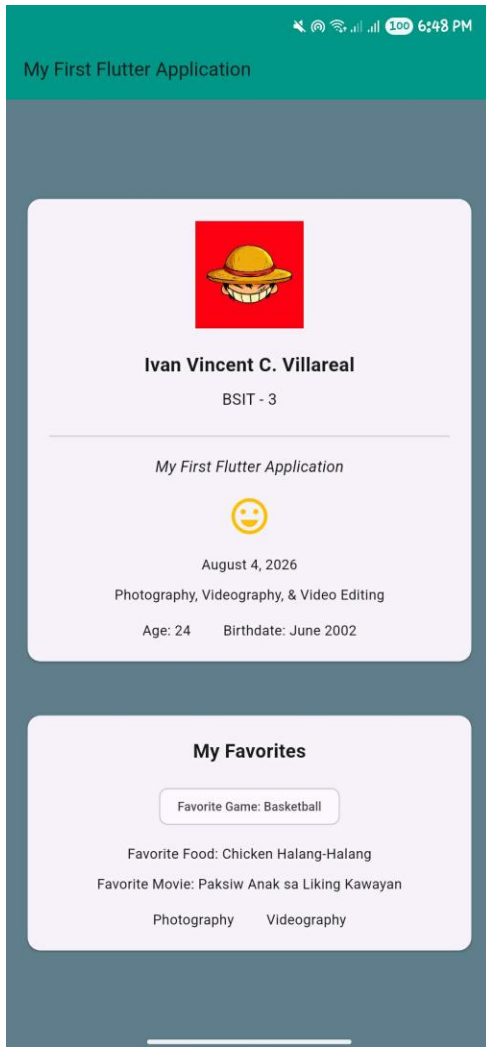
🚩 Flag 11 – Real Phone Test 📱

Objective: Run your completed application on a **physical** Android phone.

Research how to run Flutter on your phone.

Screenshot:

- Your completed application running on the physical phone (screenshot from your actual phone)



🚩 Flag 12 – Save Your Progress

Objective: Commit your completed Widget branch.

Follow proper standards of commit messages

Screenshot:

- Successful commit

- feature/widgets branch

```
johnf@GJ MINGW64 ~/Downloads/ggs (feature/widgets)
$ git commit -m "Feat: Added Main changes after done with flag 12"
[feature/widgets 52ed753] Feat: Added Main changes after done with flag 12
1 file changed, 181 insertions(+), 35 deletions(-)
```